Reconceptualizing Shadow IT: Motivations, Consequences, and Governance in the Age of Shadow AI

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Zusammenfassung

Eine Kurzzusammenfassung der Vorgehensweise und der wesentlichen Ergebnisse.
Allgemeine Merkmale

- Objektivität: Es soll sich jeder persönlichen Wertung enthalten.
- Kürze: Es soll so kurz wie möglich sein.
- Verständlichkeit: Es weist eine klare, nachvollziehbare Sprache und Struktur auf.
- Vollständigkeit: Alle wesentlichen Sachverhalte sollen enthalten sein.
- Genauigkeit: Es soll genau die Inhalte und die Meinung der Originalarbeit wiedergeben.

Abstract

Obige Zusammenfassung in englischer Sprache.

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List of Abbreviations

AI Artificial Intelligence

BYOD Bring Your Own Device

IT Information Technology

SAI Shadow AI

SIT Shadow IT

SLR Structured Literature Review

1 Introduction - 2 pages

Dieser Teil der Arbeit sollte folgende Inhalte haben: • Einführung in die Problemstellung • Motivation und Herleitung des Themas • Aufbau der Arbeit Hinweis: Es hat sich als hilfreich erwiesen, die Einleitung mit der Zusammenfassung bzw. dem Abstract und der Schlussfolgerung zu vergleichen. Damit stellt man sicher, dass diese inhaltlich im Bezug auf Zielsetzung und Motivation übereinstimmen. Der Umfang sollte ca. 5 % der gesamten Arbeit betragen. (= ca. 2 Seiten)

2 Theoretical Foundations - 6-8 pages

In order to establish a comprehensive understanding of Shadow IT (SIT) and its further, Shadow AI (SAI), this section provides an overview of the theoretical foundations and definitions relevant to these phenomena.

2.1 Shadow IT

2.1.1 Concept and Definition

SIT is a multifaceted phenomenon encompassing the use, development, or deployment of Information Technology (IT) artifacts (e.g., software, hardware, or services) outside the formal control, approval, or awareness of the central IT function. For instance, Haag and Eckhardt (2017, p. 1) describe it as "hardware, software, or services built, introduced, and/or used for the job without explicit approval or even knowledge of the organization," whereas Kopper and Westner (2016, p. 1) emphasize autonomous deployment within business departments without IT involvement. Other authors similarly highlight the lack of alignment with IT governance (Käss et al., 2021, p. 4; Rakovic et al., 2020, p. 4).

Beyond these operational perspectives, SIT can be conceptualized as a structural complement to formal IT, consisting of independently developed and used systems, processes, and practices that are not known, supported, or accepted by the official IT department (Rakovic et al., 2020, p. 147). More concise definitions further highlight either the user-driven development outside central IT or the absence of approval and monitoring as defining features (Rakovic et al., 2020, p. 4; Magunduni and Chigona, n.d., p. 2).

Collectively, these definitions establish SIT as a theoretically grounded construct characterized by autonomy, decentralization, and non-sanctioned IT activity, emerging outside formal IT governance.

2.1.2 Scope

SIT encompasses a wide range of tools and solutions, including spreadsheets, databases, cloud services, mobile devices, and other ad-hoc IT solutions (Rakovic et al., 2020, p. 2; Kopper and Westner, 2016, p. 1). These systems are typically small-scale, developed by individuals or small teams, and tailored to immediate operational needs (Rakovic et al., 2020, p. 2). Their emergence reflects both the accessibility of modern IT tools and the flexibility required by employees in dynamic work environments.

2.1.3 Conceptual Distinctions

While SIT shares similarities with related practices such as workarounds, and Bring Your Own Device (BYOD) programs, it is distinct in being unauthorized and unsanctioned. Workarounds represent behavioral adaptations to organizational constraints, whereas SIT manifests these adaptations through technological artifacts. IT consumerization, on the other hand, refers to the broader trend of user-driven technology adoption, providing a context in which SIT can emerge (Rakovic et al., 2020, p. 2; Haag and Eckhardt, 2017, p. 2).

2.1.4 Core Characteristics

SIT is not necessarily malicious or risky in intention. Users often deploy it to enhance productivity, overcome system limitations, or facilitate collaboration. Its defining features are low visibility, lack of IT support, and a tension between organizational control and individual initiative.

2.2 Shadow AI

Definition of Shadow AI

How Shadow AI differs from Shadow IT

Unique Characteristics

Examples

Emergence and Growth

Scale and Prevalence

Sociotechnical Nature and Governance Challenges

Knowledge Leakage and Data Risk

Governance Drift and Policy Gaps

Organizational Context

Attack Surface and Technical Risks

Ethical Paradox

2.3 Conceptual Background on Shadow Behavior

This section should: • briefly explain why shadow behavior exists in general • stay high-level and theory-driven • avoid listing concrete motivations

Example content: • Workarounds in organizations • Tension between control and autonomy • Informal vs. formal IT usage • Knowledge worker behavior

3 Methodology - 2-3 pages

The review follows a structured approach based on the guidelines provided by Okoli (2015, pp. 887–902) for conducting a Structured Literature Review (SLR) in the field of information systems research. The methodology is organized into four phases of two steps each: Planning, Selection, Extraction, and Execution (Okoli, 2015, p. 885).

While the original framework is designed for collaborative research settings involving multiple reviewers, this review was conducted by a single researcher. Therefore, minor adaptions were made to certain steps (e.g., drafting the research protocol, team training) to ensure methodological rigor while maintaining the integrity of the review process. The following sections describe these phases and their implementation in the context of this review.

3.1 Purpose

The first step in conducting a SLR is to define the purpose of the review, which is to identify and analyze existing literature on the phenomenon of SIT and further the development of SAI in organizational contexts. The SLR aims to synthesize insights on the motivations behind SIT and SAI, their organizational consequences, and the governance approaches employed to manage these phenomena.

3.2 Draft Protocol

The second step includes drafting a research question, a research protocol, and training the team. Since this thesis is conducted by a single person, the protocol and team training steps are not required and will therefore not be carried out further. The research questions guiding this review are as follows:

Main Research Question (RQ): How does SAI differ from traditional SIT in organizations with regard to motivational factors, consequences, and countermeasures?

To better address the research question, it will be broken down into three sub-questions and answered throughout the paper:

1. Which motivational factors known from SIT are transferable to SAI, and which additional motivation factors are specific to SAI?
2. Which organizational consequences of SAI differ from those of SIT?

3. Which countermeasures are discussed for SIT and SAI, and how do they differ in design and expected effectiveness?

3.3 Practical Screening

To make sure only relevant and high-quality literature is included in the review, practical screening criteria are defined to filter the large number of initial search results before reviewing them in detail.

For this review, only journal articles, conference proceedings, and book chapters/sections published in English between 2005 and 2026 with a focus on SIT/SAI in organizational or professional settings were included. The research design of the included studies could be empirical (quantitative, qualitative, or mixed methods), theoretical, conceptual, or review-based, as long as they provide insights relevant to the research questions. The participants in the included studies could be professionals, managers, employees in various roles (e.g., analysts, data scientists, IT staff), or end-users as long as they have a close connection to the organizational context of SIT and SAI.

3.4 Literature Search

The literature search was conducted using four search strings across four databases (AISeL, EBSCO, IEEE, and Web of Science), employing advanced search with Boolean operators (`OR` and `AND`) applied to all fields or title/abstract/keywords depending on the database. The search strings were designed to capture literature on both concepts, as well as their intersection, to ensure a broad and comprehensive coverage of relevant studies. The search strings and their purposes are summarized in Table 1.

Table 1: Literature Search Strings

String	**Purpose**	**Query**
S1	Shadow IT Baseline	`"shadow IT" OR "rogue IT" OR "IT workaround" OR "feral system"`
S2	Shadow AI Baseline	`"shadow AI" OR "unauthorized AI" OR "unsanctioned AI" OR "BYOAI"`
S3.1	Shadow IT × Shadow AI	`("shadow IT" OR "rogue IT" OR "IT workaround" OR "feral system") AND ("shadow AI" OR "unauthorized AI" OR "unsanctioned AI" OR "BYOAI")`
S3.2	Shadow IT × AI (broader)	`"shadow IT" AND "artificial intelligence"`

String S3.2 was included to capture literature not explicitly using the term *Shadow AI* but discussing the intersection of SIT and Artificial Intelligence (AI) technologies.

Additionally, the search was extended by backward and forward searching of the reference list of the included studies to identify any relevant literature that may have been missed in the initial database search.

3.5 Data Extraction

The remaining literature was subjected to a systematic data extraction process, where the literature was coded and sorted based on their key scope in accordance to the research questions. Based on this preliminary coding, relevant information was extracted and collected in a structured format to group similar concepts and themes together. This process was conducted iteratively, with multiple rounds of coding and extraction to ensure that all relevant information was captured and organized effectively for further analysis and synthesis in the subsequent phases of the review.

3.6 Quality Appraisal

While the practical screening criteria help filter out irrelevant studies before even reading the full text, the review process also involves a quality appraisal to ensure the included studies meet certain standards. Therefore exclusion criteria need to be defined and applied to the extracted studies (Okoli, 2015, p. 884).

This SLR excluded studies with unsufficient theoretical contribution, unclear methodology or findings, or low prerequisites to adequately answer the research questions.

3.7 Study Synthesization

Study synthesization involves aggregating, summarizing, and interpreting the findings from the included studies to answer the research questions (Okoli, 2015, p. 899). The aggregation utilized the preliminary outline of the review, which was based on the research questions, to group similar concepts and themes, such as definitions, motivations, consequences, and countermeasures of SIT and SAI. [1]

3.8 Writing the Review

The final phase of the review process presents the results in detail, so future reviewers can understand and reproduce the process and the findings (Okoli, 2015, p. 901). Section 4 presents the results of the SLR, organized around the key themes of identified literature, motivational factors, organizational consequences, and governance approaches of SIT and SAI to summarize existing literature and knowledge about these concepts.

[1] To support the aggregation and summarization of relevant data, generative AI tools such as ChatGPT (OpenAI, GPT-5.3, April 2026 version) and Claude (Antrophic, Sonnet 4.5, April 2026 version) were used. All outputs were critically reviewed and revised by the author who assumes full responsibility for the final content.

4 Structured Literature Review Results - 8-10 pages

4.1 Overview of Identified Literature

Table 2: Database Search Results per String

String	AISeL	EBSCO	IEEE	WoS	Total
S1	296	227	39	94	656
S2	6	79	9	5	99
S3.1	4	2	0	0	6
S3.2	45	18	4	3	70
Total	**351**	**326**	**52**	**102**	**831**

After applying the prior steps and the exclusion criteria, Table 3 summarizes the final set of included studies for the review:

Table 3: Literature Screening Process

Stage	Included	Excluded	Remaining
Records identified	–	–	831
After duplicate removal	–	124	707
Practical screening			
Language (English only)	–	24	683
Publication year (2005–present)	–	30	653
Publication type	–	15	638
Content screening			
Title screening	254	384	254
Abstract screening	122	132	122
Full-text review			
Not accessible	–	19	103
Full-text screening	62	41	62
Forward searching	3	–	65
Backward searching	4	–	69

The quality appraisal of the full texts resulted in a final set of 69 studies, which were included in the review and subsequently analyzed and synthesized to answer the research questions.

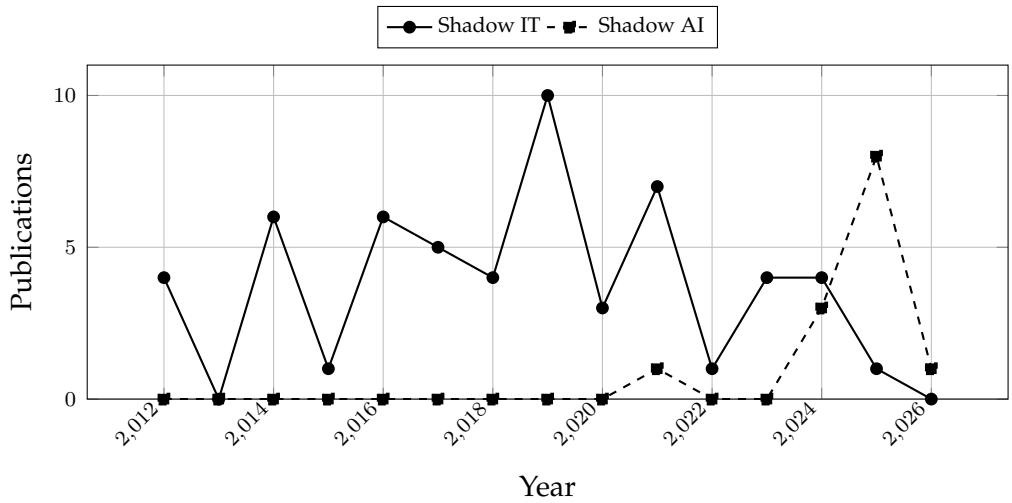

Figure 1: Development of publications on Shadow IT and Shadow AI over time

4.2 Motivational Factors

4.2.1 Individual Motivations

User Experience and Ease of Use Percieved Usefulness Performance Expectancy Effort Expectancy

4.2.2 Social Motivations

Social Influence Social Presence Subjective Norms

4.2.3 Organizational Motivations

IT Department Limitations System Inadequacies and Business-IT Misalignment Facilitating Conditions

4.2.4 Technological Motivations

Consumer IT Capabilities and Relative Advantage

4.2.5 Work-Related Stressors

Work Overload Work Pressure

4.3 Organizational Consequences

4.3.1 Positive Consequences

Innovation Benefits Individual Performance Improvement Knowledge Sharing Enhancement Productivity Gains Agility and Flexibility

4.3.2 Negative Consequences

Security Risks and Data Breaches Compliance Violations Data Leakage Governance Challenges and Integration Issues Lack of Awareness

4.4 Governance Approaches and Countermeasures

4.4.1 IT Governance Strategies

Control-Oriented Approach User-Oriented Approach Middle Ground/Balanced Approach

4.4.2 Policy Approaches

Communication and Awareness Understanding and Procedural Justice

4.4.3 Management Strategies

Balancing User Needs and Security Manager and Leadership Influence

4.4.4 Risk Mitigation & Compliance Mechanisms

Deterrence and Sanctions Personal and ORganizational Ethics Cultural Factors

5 Conceptual Development: Extending Shadow IT to Shadow AI - 4-5 pages

5.1 Transferable Motivations

Performance Improvement/Performance Expectancy Ease of Use/Effort Expectancy/User-Friendliness Autonomy Needs/Independence/Self-Determination Productivity Gains/Efficiency Social Influence/Peer Influence/Normative Beliefs Workaround Behavior/Overcoming Constraints System Limitations/IT Department Limitations

5.2 Novel Motivations

AI Capabilities (NLP, Content Generation) Accessibility of AI Tools Generative Capabilities Advanced Analytical Features Speed of AI Responses/Rapid Deployment Conversational Interfaces/Natural Language Interaction AI as Knowledge Assistant/Cognitive Support Unique Capabilites

5.3 Behavioral Shift in Knowledge Work

Changes in Knowledge Work Patterns/Knowledge Work Transformation Shift in Employee-Technology Interaction/Human-AI Interaction AI as Cognitive Assistant/Thinking Partner Transformation of Decision-Making Processes Changes in Information Processing/Cognitive Processes Impact on Professional Expertise/Expert Knowledge Changes in Work Practices/Work Routines

5.4 Integrated Conceptual Framework

Conceptual models connecting Motivations -> Behaviors -> Outcomes Theoretical Frameworks for understanding Shadow IT/AI Relationships between Constructs Mediating Factors/Mediation Relationships Moderating Factors/Moderation Effects Causal Pathways/Causal Chains Integration of Mulitple Theories Organizational-Individual Factor Integration Sociotechnical Integration (Shadow AI)

6 Comparative Analysis of Shadow IT and Shadow AI - 3-4 pages

6.1 Differences in Motivational Drivers

6.2 Differences in Organizational Risk

6.3 Differences in Governance Approaches

7 Discussion - 3-4 pages

8 Limitations and Future Research Directions - 1-2 pages

9 Conclusion - 2 pages

In der Schlussfolgerung sollen

• die Themenstellung • der gewählte Ansatz • die Ergebnisse der Arbeit • eine kritische Stellungnahme/Einschätzung • nächste Schritte

deutlich werden. Hinweis: Die Schlussfolgerung sollte mit der Zusammenfassung bzw. dem Abstract und der Einleitung abgeglichen werden. Es sollte immer eine Zusammenfassung der wesentlichen Erkenntnisse der eigenen Arbeit sein, die den Forschungsbeitrag darstellt. Der Umfang der Schlussfolgerung sollte ähnlich wie die Einleitung ca. 5 % der gesamten Arbeit betragen. (= ca. 2 Seiten)

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A Appendix A